Assignment 5

1. The irrational number π can be approximated by the following series

   (a) $\pi = \frac{6}{\sqrt{3}} \sum_{k=0}^{\infty} \frac{(-1)^k}{3^k\,(2k+1)}$

   (b) $\pi = \sqrt{12} \sum_{k=0}^{\infty} \frac{(-3)^{-k}}{2k+1}$

   (c) $\pi = 4 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)}$

   Write functions pi_1(n), pi_2(n), pi_3(n) that approximates π using finite versions of the summations above from 0 to n. Submit only your functions as a .py file.